

TITLE

Analysis of Correlation between Hand Anthropometry and Hand Grip Strength



Human Factors Engineering

Course Code: IE-317

Final Project Report

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1. Introduction:

The hand is the part of the upper extremity from the radiocarpal joint to the end of the fingers and has a key role in terms of all upper extremity functions. The anatomic structure of the hand provides the hand with both fine and coarse grip motion through interactions that are sometimes agonist and sometimes antagonist. Therefore, the anatomy of the hand should be studied as a whole including the skin, tendons, muscles, bones, joints, vessels and nerves (Gövsa, 2008). Complex movements of the hand are formed by means of good coordination between the balanced muscle system of the hand and the central nervous system (Snell, 2011). The thumb is more important functionally compared to the other fingers. Functional loss of the thumb results in a 40% decrease in the hold and grip functions (Gövsa, 2008). Among the functions of the hand, handgrip strength is important for the execution of daily activities. Grip strength of the hand is defined as the voluntary flexion of all fingers of the hand in maximum power under normal bio-kinetic circumstances (Fallahi and Jadidian, 2011). It has been accepted as an objective measure of upper extremity performance in addition to being a reliable marker that provides general information on a person's nutritional status, physical power and general health status. A hand's grip strength is a physiological variable affected by age, sex and bodily dimensions (Jurimae et al., 2009; Koley and Pal Kaur, 2011). Knowing the functional structure of the hand and handgrip strength is important for surgical approaches to the hand, postsurgical performance evaluations and clinical follow-up in patients with upper extremity injuries or undergoing hand rehabilitation. Handgrip strength is not only a good predictor for evaluating the efficiency of a treatment procedure and the rehabilitation process of hand disorders, it is also a good predictor for neuromuscular and chronic metabolic diseases (Fallahi and Jadidian, 2011; Otto et al., 2014). Handgrip strength is also an important and required parameter in the evaluation of performance in some sports such as tennis, basketball, volleyball, athletics and wrestling. Previous studies have so far only been made on healthy children or on different sports groups and studies on the reference values of handgrip strength in young adults are limited. On the other hand, studies on various sports groups have demonstrated the effect of hand dimensions on handgrip strength in athletes who use their hands for grasping (Fallahi and Jadidian, 2011). We hypothesized that maximum handgrip strength is largely related to the anthropometric parameters of the hand, in particular, the distance between the distal wrist line and the distal end of the fingers. Increased finger length will increase the surface of the hand and therefore increase handgrip strength. The aim of this study was to evaluate handgrip strength in healthy young adults and its association with upper extremity anthropometric measurements. The results of the present study may lead the investigators to study the correlation between handgrip strength and upper extremity dimensions in various sports groups as well.

- ▶ Understanding the relationship between hand anthropometry involves measuring the dimensions of the hand, and hand grip strength is crucial in various fields such as ergonomics.
- ▶ Hand grip strength is a reliable indicator of overall muscle strength and functional ability, while hand anthropometry provides insights into hand structure and function.

- ▶ Exploring the correlation between these two factors can offer valuable information for improving ergonomic designs, and enhancing physical performance.
- ▶ This analysis aims to investigate the relationship between hand anthropometry and hand grip strength to deepen our understanding of hand function and its implications for human performance and well-being.

2. Literature Review:

2.1. Human Factors Engineering:

Human factors engineering, also known as ergonomics or human factors psychology, is a multidisciplinary field that focuses on understanding human capabilities and limitations and applying that knowledge to the design of systems, products, and environments to optimize human well-being and performance.

Human factors engineering considers various aspects of human behavior, cognition, physiology, and biomechanics to ensure that systems are designed to be usable, safe, efficient, and comfortable for the people who interact with them. This field draws on principles from psychology, engineering, physiology, biomechanics, anthropology, and other disciplines.

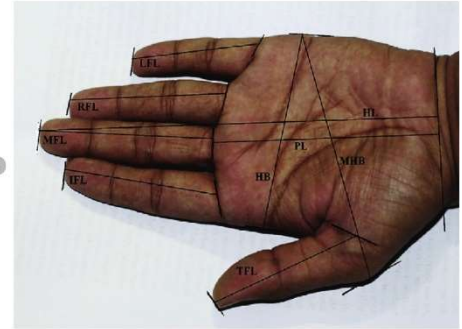
Applications of human factors engineering can be found in a wide range of domains, including:

- 2.1.1. **Product design:** Designing consumer products, tools, and equipment to be intuitive, comfortable, and safe for users.
- 2.1.2. **Workplace design:** Designing workspaces, equipment layouts, and workflows to minimize physical strain, fatigue, and errors.
- 2.1.3. **Transportation:** Designing vehicle controls, cockpit layouts, and transportation systems to optimize safety and usability for drivers, pilots, and passengers.
- 2.1.4. **Healthcare:** Designing medical devices, interfaces, and healthcare environments to enhance patient safety, comfort, and efficiency of care delivery.
- 2.1.5. **User interface design:** Designing interfaces for software, websites, and digital devices to be intuitive, user-friendly, and accessible to diverse user populations.
- 2.1.6. **Safety systems:** Designing safety-critical systems, such as nuclear power plants or air traffic control systems, to prevent errors and minimize the consequences of human mistakes.

Overall, human factors engineering aims to create systems that are well-adapted to the capabilities and limitations of the people who use them, ultimately improving usability, safety, efficiency, and overall user experience.

2.2. Hand Anthropometry:

Hand anthropometry is the scientific study of the measurements and proportions of the human hand. It involves analyzing various dimensions of the hand, such as length, width, circumference, and proportions of different segments like fingers and palm. Hand anthropometry is essential in various fields, including ergonomics, product design, glove design, and forensic science. By understanding the dimensions and proportions of the human hand, designers can create products that are comfortable, functional, and safe for users.



Hand anthropometric measurements. HL: Hand Length, HB: Hand Breadth, MHB: Maximum Hand Breadth, PL: Palm Length, TFL: Thumb Length, RFL: Index Finger Length, MFL: Middle Finger Length, RFL: Ring Finger Length and LFL: Little Finger Length.

2.3. Hand Grip Streangth:

Hand grip strength is a measure of the force generated by the muscles in your hand and forearm when you squeeze an object. It's commonly assessed using a hand dynamometer, which records the amount of force exerted. This measurement provides insights into overall muscle strength and function, as well as potential issues such as weakness or injury. Hand grip strength is important for various activities in daily life, sports performance, and occupational tasks that require grasping and holding objects securely.

Grip strength of the hand is defined as the voluntary flexion of all fingers in maximum strength under normal bio-kinetic circumstances.

2.4. Grip Strength Measurment:

Handgrip strength is an important parameter in terms of the efficient use of the hand in addition to being a reliable marker of the physical strength of the individual. The aim of this study was to evaluate the association between the anthropometric measurements of the hand and handgrip strength. A total of 150 students from Baskent University aged 18 to 25 years participated in this study. The participants' height, weight, length of upper extremity, arm, forearm and hand, arm circumference, wrist circumference, elbow and wrist width and hand width were measured in this study. In addition, the distance from the distal wrist line to the distal end of each finger and the distances from each fingertip to the peak point of the thumb were measured with the fingers in abduction. The measurements for the fingers were obtained from scanned images of the hands. Handgrip strength, however, was measured using a hand dynamometer. The handgrip strength of the males was found to be greater compared to the females. A statistically significant association between handgrip strength and all upper extremity parameters was found. The most significant of all the parameters was the distance between the wrist and thumb of the right and left hand (WT). This study is a preliminary one and may be seen in terms of treatment approaches to the hand, clinical follow-up, and hand rehabilitation.



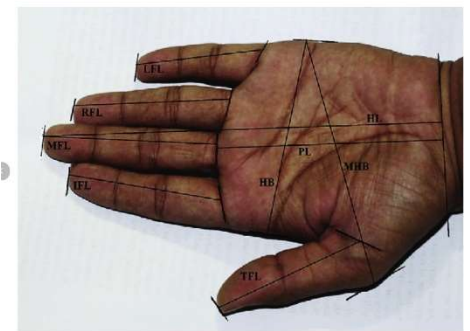
Hand grip strength is measured using a handheld dynamometer, a device designed for this purpose. The individual holds the dynamometer in their hand and squeezes the handle with maximum force. The dynamometer then records the amount of force exerted by the hand muscles, usually displayed in kilograms or pounds. This measurement offers valuable insights into overall muscle strength and function, useful in various contexts such as occupational health.

The grip strength of all the participants is measured at three different levels to check how increase in distance effect the grip strength.

3. Methodology:

The method that is implemented to measure the correlation between Hand Anthropometry and Grip Strength mainly comprises of Following Steps:

1. Firstly we gathered the apparatus in order to measure Hand Anthropometry and Grip strength.
2. So those apparatus mainly named as Measuring Tape for Hand Anthropometry and Hand Dynamometer for Grip Strength.
3. Almost 50 University Students Measurement were taken.
4. So the data that was taken through Measuring tape for Hand Anthropometry mainly comprises of:
 - Height
 - Hand Length
 - Hand Breadth
 - Max. Hand Breadth
 - Palm Length
 - Thumb Length
 - Index Finger Length
 - Middle Finger Length
 - Ring Finger Length
 - Little Finger Length
5. Through Hand Dynamometer Grip Strength was taken of Right Hand in stages of Easy, Medium and Hard mainly named as Level 1, Level 2 and Level 3. So these levels mainly prescribes us about the power that the hand will require to generate force.
6. When all this data was taken through the apparatuses then they were added on to the excel sheet for further analysis and Computations.



Hand anthropometric measurements: HL: Hand Length, HB: Hand Breadth, MHB: Maximum Hand Breadth, PL: Palm Length, TFL: Thumb Length, IFL: Index Finger Length, MFL: Middle Finger Length, RFL: Ring Finger Length and LFL: Little Finger Length.

7. After that we have analyzed that data by calculating Maximum, Mean and Minimum values of Grip Strength and Hand Anthropometry.
8. Similarly we have also carried out the correlation between Hand Anthropometry and Grip Strength which is quite simple that when the levels of Hand Dynamometer is increasing the grip strength is decreased and it can simply be noticed through the value of correlation which is further shown in results and Discussion.
9. We have also describe the results through Bar Charts which mainly indicates that in what range the maximum, minimum and average value lies of the Grip Strength and Anthropometry.



4. Results and Discussion:

4.1. Students Reading:

Almost 50 University Students Reading was noted on Excel sheet of Hand Anthropometry and Grip Strength which can be seen below:

Personal Information				Hand Anthropometric Data										Dynamometer Readings		
Sr. No	Student Name	Reg.No	Age	Height (cm)	Hand Length (cm)	Hand Breadth (cm)	Max. Hand Breadth (cm)	Palm Length (cm)	Thumb Length (cm)	IFL (cm)	MFL (cm)	RFL (cm)	LFL (cm)	Step1 (kg)	Step2 (kg)	Step3 (kg)
1	Shaheer Ahmad	21-IE-22	22	191	21	10	12	11.5	8.5	8.2	9.1	8.1	6.7	52	46	32
2	Arslan Ahmad	21-IE-56	19	177.3	19	10	12.5	12	6	7.5	8	7	6	48	45	42
3	Iqbal Rawal	21-IE-04	22	177.5	20	10.5	13	11.5	8	8	8.5	8	6.5	52	48	43
4	Ali Sajjad	21-IE-14	21	169.8	21	8.5	11.5	11	8	8.5	9	8	6.5	38	30	28
5	Rafay Ali	22-IE-12	19	170.6	18	9	12	11	6.5	7.5	8	7.5	5.5	48	44	34
6	Ahmad	21-ENC-12	21	171.4	19	9.5	12	11	7	7	7.5	7	6	40	32	22
7	Umar Mushtaq	22-ENC-1	20	154	17.5	8.5	11	10	6	6.5	7.5	7	6	38	28	22
8	Umar Nadeem	22-ENC-37	20	159	19	9.5	12	11	7	7.5	8	7.8	6	41	40	34
9	Hamid Shah	22-ENC-33	20	172	19	9	11	11	6.5	7	8	7.5	6	42	36	33
10	Awaim	22-ENC-33	21	176	20	8	10	11	7	7.5	8.5	8	6.5	39	30	29
11	Shayan	21-EE-144	20	168.1	19.5	10.3	12	11	6.2	7.5	8.5	7.5	6	41	40	30
12	Aftab	22-ENC-35	19	166.5	18	9	11.5	10.5	6.5	6.8	7.2	7	5.5	36	31	26
13	Uzair	22-IE-24	19	193.8	21.2	10	12.5	11.5	7.3	8.5	9.5	8.3	7	55	49	42
14	Hamza Hameed	22-IE-23	20	171.6	18.8	9	12	11	6.5	7	8	7.3	5.8	42	36	26
15	Shahmeer Tiwani	23-IE-27	19	191.4	22.86	12.5	10.5	12.8	8.4	9.1	10	9	7.5	52	48	40
16	Talha	23-IE-23	20	171.3	19.3	10	11.5	11	6.5	7	8.5	7.5	6	58	50	44
17	Huzaifa	21-IE-27	22	178.6	19.5	11.3	9	10	7	6	8.5	7.5	7.6	42	37	30
18	Ghulam Ghoss Ali	23-IE-03	20	183	18.7	9.2	10	10.5	7	7.5	8.2	7.6	6	48	43	42
19	Ghufran Aleem	23-IE-04	19	160	18	8	9.5	10	7	7.8	8	7	6	46	47	43
20	Hanzala Masood	23-IE-06	19	181	19.5	9.5	11.5	11	7	7.3	8.5	7.5	6.5	53	49	49
21	Sikandar Abbas	21-IE-41	21	165	17.5	8.5	10	10.5	6.5	7	7	6.5	5.5	43	41	39
22	Monis Raza	21-EE-70	20	166	18.6	9.2	11.3	10	7	8.2	8.6	8	7.5	46	45	42
23	Hamad Hassan	21-CS-148	23	165	19.5	8.2	9.3	11.2	7.1	7.8	8.3	8	6.5	48	46	47
24	Fawad Akhtar	21-CS-149	21	172	20	8.5	9.8	11.5	7.3	8	8.5	8.2	6.7	32	33	32
25	Saad Ahmad	21-CE-143	22	191	19	8.5	9.2	10.3	7.1	7	7.7	7	6.2	53	52	49
26	Saad Ur Rehman	21-CE-100	21	176	19.5	9.5	10	10.5	8	8.5	9	8.3	7.2	34	32	33
27	Kashif Mehmood	21-EE-37	23	176	20	8.6	9.5	11.6	8.3	7.9	8.4	8	7	38	37	32
28	Abdul hanan	21-ME-144	22	163	18	8.2	9	10	6.8	7.8	8	7.5	6	30	28	24
29	Jam Muhammad Talha Laar	20-IE-07	23	182.88	18	8	16	10	7.5	8.1	9	7.5	6	59	55	52.3

30	Baseer	21-ME-159	22	165	18.5	10.5	11	11	7.5	8	8	7	6.5	53	50	48
31	Mukarram Rehman	22-IE-25	20	175.1	20	10.5	15.5	11.5	20	8.5	9.5	8	6.5	42	43	39
32	Jamaal	23-CS-182	19	168	17.5	9.5	15.5	10.5	7.5	6.5	8	7	5.5	39	37	33
33	Ermina Mughal	20-IE-23	23	157.2	16.6	7.9	14.5	9.7	5.8	6.5	6.5	6.2	5	35	33.5	28
34	Muhammad Taiha	20-ME-40	23	173.2	19	10.5	15.5	11	19	7	8.5	7.5	6.5	53	49	46
35	Muhammad Ali Raza	21-IE-21	21	174.8	18	10.5	15.5	10	8.5	7	7.5	6	5.5	40	37	34
36	Syeda Fatima Zaidi	21-IE-03	21	161.3	18.8	8	14	10	6.2	7.2	7.6	7.3	5.8	32	29	23
37	Noor-ul-Fajar	21-IE-16	20	161.2	17.8	7.6	14.5	10	6.6	7.9	7.9	7	5.3	35	31	29
38	Iraj Fatima	23-IE-24	19	158.6	16.5	7.5	13.6	9.6	5.8	6.1	6.8	5.9	4.8	33	29	27
39	Khizer Majeed	21-IE-57	21	172.6	20.2	8.6	11.5	12	7	7.5	8.5	7.5	6.5	38	30	27
40	Anas Liaqat	21-IE-34	21	164.4	18.5	9.8	12	10.5	6.2	7	8	7.7	5.8	48	38	28
41	Zain ul Abideen	21-IE-42	22	170	19.3	10	12.3	11.5	7	7	8	7.5	6.5	47	42	35
42	Sajeel Tariq	21-IE-19	21	166	20	9.5	12.5	11.5	7.5	8	9.3	7.8	6.5	45	42	37
43	Kamran Arif	21-CS-129	23	167.5	20.7	9.5	11.8	11.9	7.7	8	8.5	8	7	39	33	28
44	Ahmed Javed	21-IE-10	21	173.5	19	9	11.5	10	7.1	8	8.8	7.8	6.6	40	36	33
45	Burhan Ud Din	21-IE-09	21	167	18.2	9	11.2	10.8	7	7	8.2	7.2	6	39	33	22
46	Turab Ali	21-IE-24	22	178.3	19	9.3	10.9	11.2	6.7	7.6	8	7.4	6	49	35	33
47	Hamza	22-IE-02	21	181.4	20	9	11	11.5	7	7.8	8.3	7.5	6.5	47	40	34
48	Usman	22-IE-28	19	178.5	19	9	11.5	10.7	6.4	6.5	8	7.5	5.7	42	38	28
49	Ahmed Azeem	21-IE-36	23	169.7	18	9.7	13	11	6.7	7.5	8	7.7	6.25	50	46	40
50	Farman Ali	21-CS-101	23	174	19.5	8.4	9.1	10.5	7	8.3	9	8	7.2	52	51	49

Similarly their mean values and standard deviation was calculated as:

Mean			171.9616	19.0612	9.236	11.78	10.846	7.514	7.498	8.248	7.492	6.243	43.84	39.61	34.846
Standard Deviation			9.021404	1.188656659	0.993604036	1.86120435	0.68993049	2.562206479	0.6748	0.679	0.587572	0.619744	7.21495	7.463715	8.195449

4.2. Value of Correlation:

Co-Relation of Hand(Breadth & Length)	0.6257401
<u>HAND LENGTH:</u>	
Co-Relation of Hand Length & Step 1	0.2716843
Co-Relation of Hand Length & Step 2	0.2131805
Co-Relation of Hand Length & Step 3	0.1472151
<u>HAND BREDTH:</u>	
Co-Relation of Hand Breadth & Step 1	0.4212925
Co-Relation of Hand Breadth & Step 2	0.3854117
Co-Relation of Hand Breadth & Step 3	0.2203391
<u>PALM LENGTH:</u>	
Co-Relation of Palm Length & Step 1	0.3003859
Co-Relation of Palm Length & Step 2	0.2296543
Co-Relation of Palm Length & Step 3	0.1375665

The correlation between grip strength and hand anthropometry (the measurements and proportions of the hand) can vary depending on several factors, including age, gender, hand size, and individual characteristics.

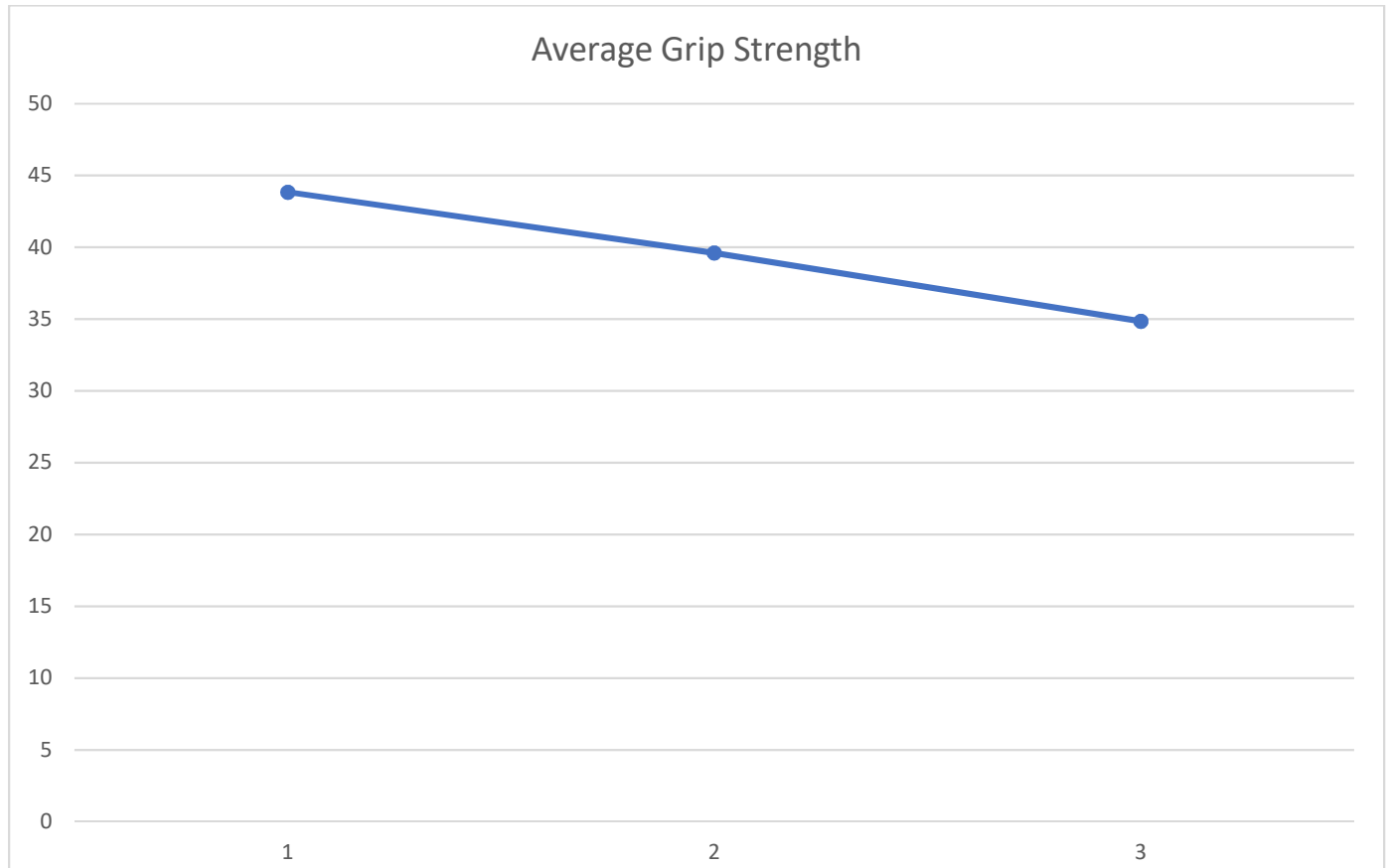
1. **Hand Size and Anthropometry:** Larger hands tend to have greater surface area and muscle mass, which can contribute to higher grip strength. However, the specific proportions of the hand, such as finger length and hand width, can also influence grip strength. For example, individuals with longer fingers or wider palms may have advantages in certain grip tasks.
2. **Muscle Mass and Anthropometry:** Anthropometric measurements of the hand, such as finger circumference and palm width, can be correlated with muscle mass and strength. Individuals with larger hand dimensions may have greater muscle mass and therefore higher grip strength, although this relationship can be influenced by factors such as body composition and physical conditioning.
3. **Gender Differences:** On average, men tend to have greater grip strength than women due to differences in muscle mass and distribution. These gender differences in grip strength can partially be attributed to differences in hand anthropometry, such as hand size and muscle fiber composition.
4. **Age:** Grip strength tends to decrease with age due to factors such as muscle atrophy and decreased flexibility. Changes in hand anthropometry, such as joint stiffness or decreased range of motion, can also contribute to declines in grip strength as individuals age.
5. **Training and Conditioning:** Grip strength can be improved through specific training exercises targeting hand muscles and grip strength. While hand anthropometry may influence baseline grip strength, individuals can enhance their grip strength through targeted training regardless of their initial hand dimensions.

Overall, while there may be correlations between grip strength and hand anthropometry, individual differences in factors such as muscle mass, training history, and physiological characteristics can influence grip strength independently of hand size or proportions.

4.3. MIN, MAX and AVG Values of Student Data:

VARIABLE	MINIMUM	MAXIMUM	MEAN	STANDARD DEVIATION
Height	154	193.8	171.9616	9.021403882
Hand Length	16.5	22.86	19.0612	1.188656659
Hand Breadth	7.5	12.5	9.236	0.993604036
Max. Hand Breadth	9	16	11.78	1.86120435
Palm Length	9.6	12.8	10.846	0.68993049
Thumb Length	5.8	8.5	7.514	2.562206479
IFL	6	9.1	7.498	0.674760729
MFL	6.5	9.1	8.248	0.678906684
RFL	5.9	9	7.492	0.587571967
LFL	4.8	7.6	6.243	0.619744022

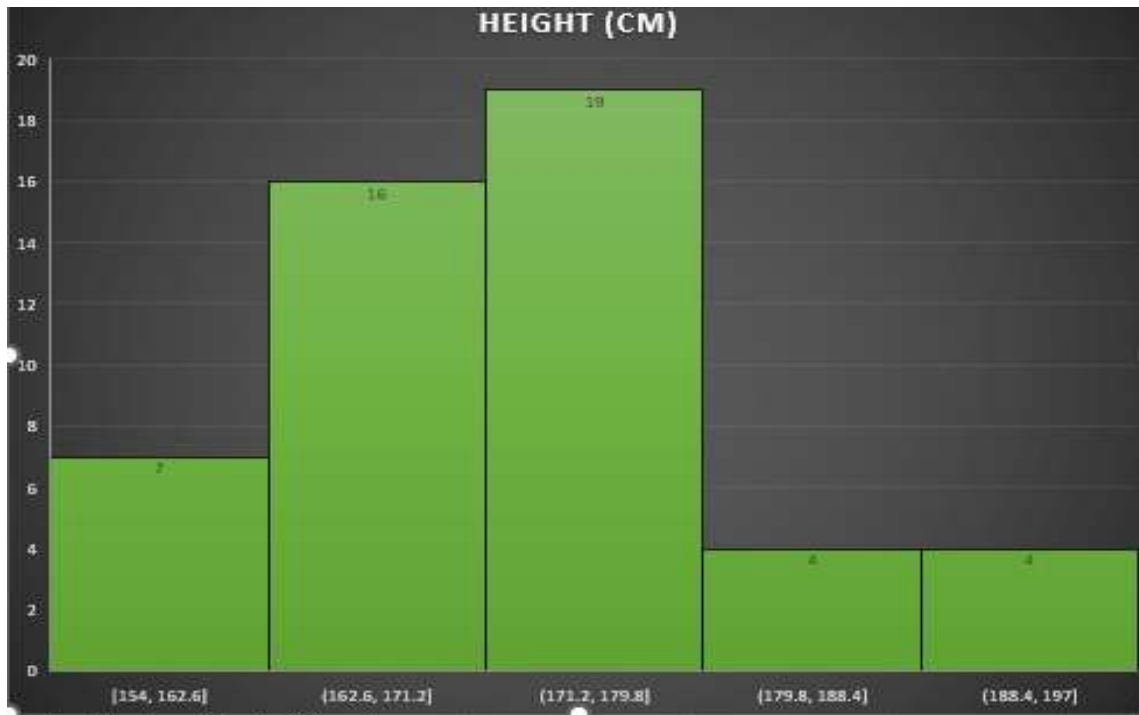
4.4. Grip Strength Rating For Males and their Graphical Representation:



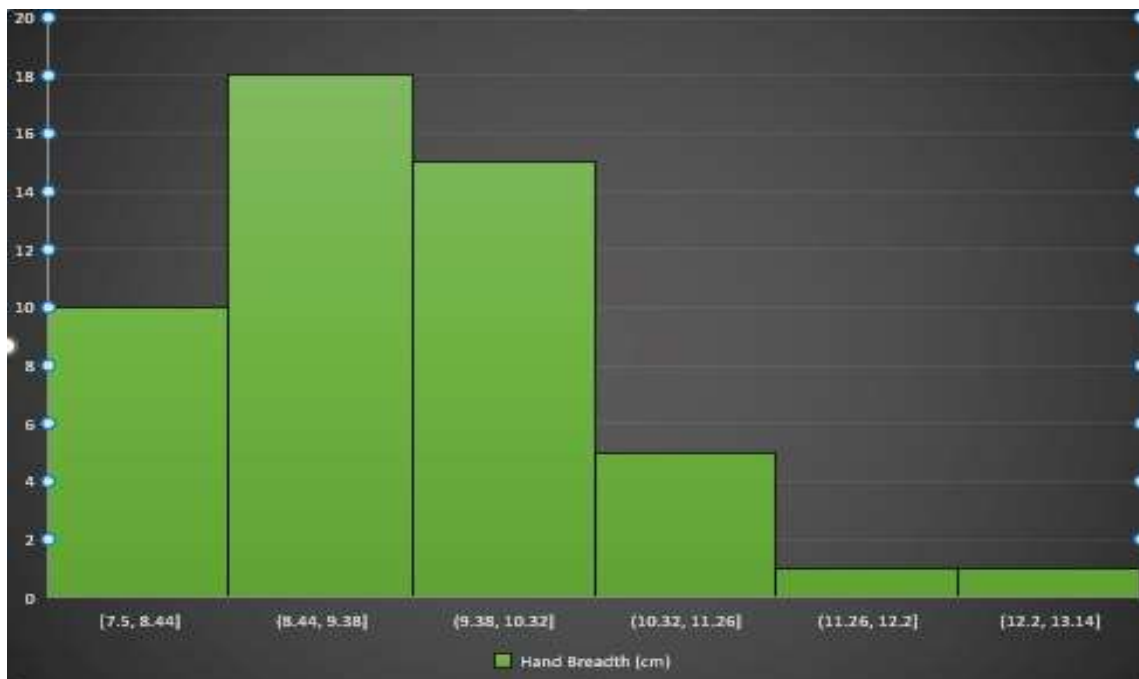
4.4.1. Grip Strength rating for Males:

AGE	Weak	Normal	Strong
10-11	< 12.6	12.6-22.4	> 22.4
12-13	< 19.4	19.4-31.2	> 31.2
14-15	< 28.5	28.5-44.3	> 44.3
16-17	< 32.6	32.6-52.4	> 52.4
18-19	< 35.7	35.7-55.5	> 55.5
20-24	< 36.8	36.8-56.6	> 56.6
25-29	< 37.7	37.7-57.5	> 57.5
30-34	< 36.0	36.0-55.8	> 55.8
35-39	< 35.8	35.8-55.6	> 55.6
40-44	< 35.5	35.5-55.3	> 55.3
45-49	< 34.7	34.7-54.5	> 54.5
50-54	< 32.9	32.9-50.7	> 50.7
55-59	< 30.7	30.7-48.5	> 48.5
60-64	< 30.2	30.2-48.0	> 48.0
65-69	< 28.2	28.2-44.0	> 44.0
70-99	< 21.3	21.3-35.1	> 35.1

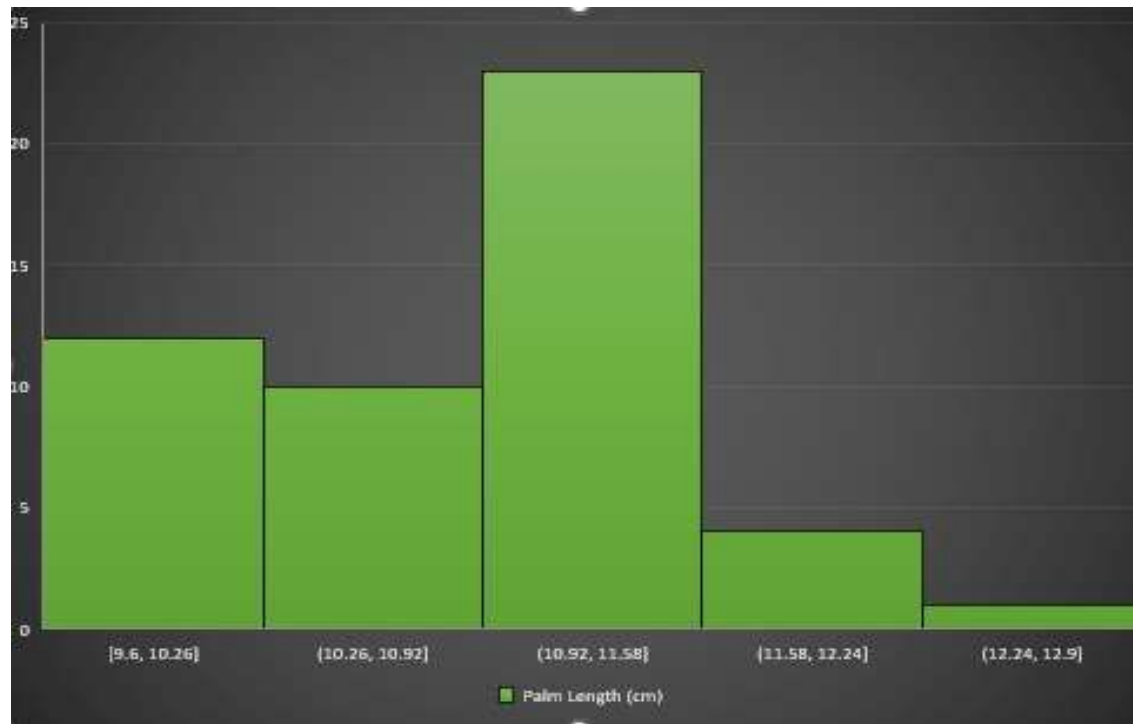
4.4.2. Height:



4.4.3. Hand Breadth:



4.4.4. Palm Length:

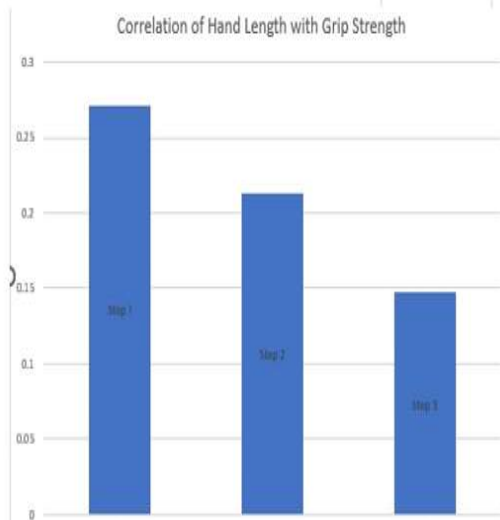


4.4.5. Hand Length:

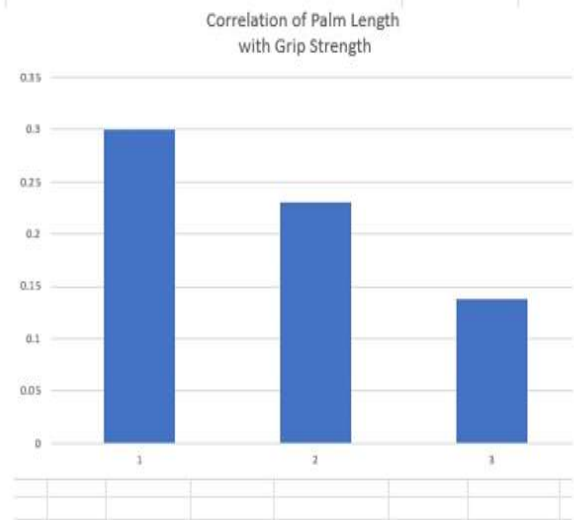


4.4.6. Correlation of Height, Hand Length, Hand Breadth & Height:

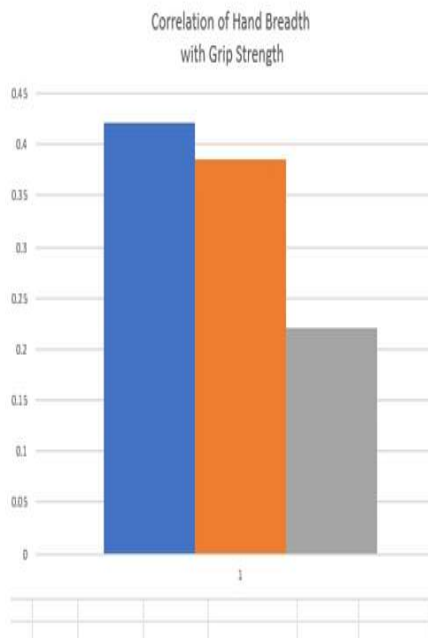
Co-Relation of Hand Length & Step 1	0.271684
Co-Relation of Hand Length & Step 2	0.213181
Co-Relation of Hand Length & Step 3	0.147215



Co-Relation of Palm Length & Step 1	0.300386
Co-Relation of Palm Length & Step 2	0.229654
Co-Relation of Palm Length & Step 3	0.137566



Co-Relation of Hand Breadth & Step 1	0.421292
Co-Relation of Hand Breadth & Step 2	0.385412
Co-Relation of Hand Breadth & Step 3	0.220339



Co-Relation of Height & Hand Length	0.62574005
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4.5. Conclusion:

It can be concluded that as the distance of the handle position on the dynamometer increases, grip strength consistently decreases across three different levels. These findings emphasize the importance of understanding how variations in handle position and distance can affect grip strength. By recognizing this relationship, we can further refine rehabilitation approaches aimed at improving grip strength and optimizing hand function without necessarily designing new tools or equipment.

4.6. References:

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